

Github Repository- <https://github.com/asad1811/prompt-dash>

Task2

Admin Dashboard url- <https://prompt-dash.onrender.com/admin.html>

User Dashboard url- <https://prompt-dash.onrender.com>

It has been deployed on Render as a single service and not two different services, the individual url of each dashboard is listed above. **It will not be functional since API key has not been provided during deployment on Render. Although the full functionality has been checked on the local system with the API key. The video demo link is attached as follows-**

<https://drive.google.com/file/d/1ecc6CqMGVv7tEK3cRro4h3XBeW3yInZa/view?usp=sharing>

The code for the following can be found under Task2 in the github repository. Admin Dashboard has a basic password protection. The user dashboard has a plain UI where the user can give a star rating and submit reviews. Validation has been applied so that both fields are necessary for submission. After successful submission the AI model delivers an appropriate response and greet. Simultaneously on the Admin Dashboard, the data is refreshed every 5 seconds. The admin dashboard has useful analytics like average star rating, count of each stars and number of feedbacks. The table has columns like Timestamp, Rating, User Review, AI summary, AI recommendations.

Admin Dashboard

Auto-refresh: 5s

Total Feedback

4

Average Rating

3.25 ★

Ratings Breakdown

1★: 0

2★: 1

3★: 1

4★: 2

5★: 0

Time	Rating	User Review	AI Summary	AI Recommended Actions
06/12/2025, 07:21:52	2 ★	ThE event was kind of boring. Although I think the story was fine but just another example of poor execution.	The user found the event boring and criticized its execution despite a decent story.	• Conduct a survey to gather more detailed feedback on event execution. • Review the event planning process to identify areas for improvement. • Consider incorporating more engaging elements or interactive activities in future events.
06/12/2025, 04:45:22	4 ★	The story was slightly off in the beginning with a blurred buildup. Although the dots did connect well in the end and I had a good time.	The user found the story's beginning unclear but enjoyed the overall experience as it connected well by the end.	• Analyze user feedback for common themes regarding story structure. • Consider revising the beginning of the story for clarity in future updates. • Gather more user reviews to assess if this is a widespread issue. • Engage with users to understand their expectations for story buildup.

Share Your Feedback

Your Rating:



Your Review:

Tell us what you liked or what we can improve...

Submit Feedback

Thank you for your feedback!

AI Response

Thank you for your thoughtful review! We're glad to hear that the movie picked up and delivered a thrilling ending. We appreciate your feedback and hope you enjoy our future releases!

Task 1

The notebook can be found under Task1 in the github repository along with the 200 Yelp sample dataset and the results for the 4 prompt versions

Dataset

We used **200 Yelp reviews**, randomly sampled from the dataset. Only two fields were retained: 1-Text(full review content) , 2.Stars- 1 to 5 rating

Model

We employed the **GPT-4.1-mini** model

Prompt Design & Iterative Improvements

We created **four** prompting strategies, increasing in structure and reasoning support:

Version	Prompt Type	Design Intent
V1 – Basic	Minimal instruction sentiment classification	Establish baseline accuracy
V2 – Strict JSON	Strong output constraints	Improve structure & reduce format errors
V3 – Rubric + Examples	Provide rating scale + few-shot examples	Guide more precise rating interpretation
V4 – Step-based Reasoning	Analyze sentiment/complaints/enthusiasm	Increase consistency via reasoning

Evaluation Metrics

Accuracy- % predictions matching true star rating

JSON Validity Rate- % responses parsed successfully

Mean Absolute Error (MAE)- Avg. difference between predicted and actual stars

System Behavior Observation

- All prompts achieved **100% valid JSON**, indicating robust structural constraints.
- More complex prompts **reduced** performance due to *over-interpretation* of simple reviews.
- The Yelp dataset's **positive skew** makes borderline cases between 4★ and 5★ most error-prone.

Result Evaluation

All versions maintained 100% valid JSON output, demonstrating excellent format reliability. However, prediction accuracy varied. Results show that simpler prompts with strong output constraints (V2) outperform richer prompts including rating rubrics or chain-of-thought reasoning. This occurs because the Yelp dataset has predominantly polarized sentiment,

making complex reasoning more prone to error through overinterpretation. Our results clearly show that **prompting is not “more accurate” simply because it is more elaborate.** Therefore, prompt engineering must balance structure and simplicity, emphasizing JSON reliability and direct sentiment cues rather than detailed factor analysis. This supports a key principle: Prompt complexity must match data complexity. Elaborate reasoning is harmful when sentiment is obvious.

Prompt Version	Strategy Type	Accuracy	MAE	JSON Validity
V1	Basic sentiment instruction	0.655	0.360	1.0
V2	Strict JSON constraints	0.660 (best)	0.355 (best)	1.0
V3	Rating rubric + few-shot examples	0.640	0.375	1.0
V4	Step-based reasoning	0.630	0.385	1.0